

KEITH, Australia

WMO: 958150

Lat: 36.1058S

Lon: 140.3272E

Elev: 90

StdP: 14.65

Time Zone: 9.50 (AUA)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.9	34.1	25.8	19.8	75.6	29.0	23.0	64.3	27.2	57.9	24.5	55.8	2.5	0	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.0	101.9	65.8	96.9	64.5	92.5	63.6	69.8	85.9	67.8	86.0	66.1	85.0	12.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.8	72.3	62.9	86.2	71.0	60.5	78.9	69.7	33.8	84.9	32.2	86.2	30.9	85.4	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	21.2	19.0	DB	28.2	110.8	1.5	3.3	27.1	113.2	26.2	115.2	25.4	117.0	24.4	119.4	(4)
(5)			WB	27.7	72.8	1.5	2.0	26.7	74.2	25.8	75.4	25.0	76.5	23.9	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	70.1	69.3	65.8	60.0	54.6	50.2	49.6	50.0	52.5	57.2	62.7	66.8	(6)	
	DBStd	9.97	8.35	7.86	7.81	6.45	5.10	4.15	3.85	4.40	5.33	7.59	8.30	8.20	(7)	
	HDD50	209	0	0	0	1	13	48	56	54	25	11	1	0	(8)	
	HDD65	2906	32	36	83	180	324	444	477	465	376	270	146	73	(9)	
	CDD50	3500	624	540	490	301	156	54	44	54	101	233	381	522	(10)	
	CDD65	724	191	157	109	30	2	0	0	0	2	27	76	130	(11)	
	CDH74	11495	3020	2249	1569	429	17	0	0	3	51	571	1375	2211	(12)	
	CDH80	5933	1705	1187	757	147	1	0	0	0	7	236	689	1204	(13)	
(14) Wind	WSAvg	8.9	9.7	9.1	8.6	7.9	8.2	7.9	9.4	9.0	8.9	9.2	9.4	9.8	(14)	
(15) Precipitation	PrecAvg	18.5	0.8	0.7	0.8	1.3	2.0	2.2	2.2	2.4	2.1	1.7	1.3	1.0	(15)	
	PrecMax	25.2	3.4	2.5	5.0	3.6	5.4	4.0	4.4	5.2	4.7	6.0	3.3	2.9	(16)	
	PrecMin	9.8	0.1	0.0	0.0	0.0	0.3	0.2	0.7	0.6	0.4	0.2	0.1	0.1	(17)	
	PrecStd	3.8	0.7	0.7	0.9	0.9	1.1	1.0	0.9	1.1	0.9	1.2	0.8	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.6	103.5	100.8	91.9	76.8	66.2	65.0	72.0	80.0	94.2	99.9	105.3	(19)	
		MCWB	68.2	68.8	65.2	63.2	57.4	54.9	54.9	56.8	58.5	61.0	64.8	66.4	(20)	
	2%	DB	103.2	97.8	93.9	83.6	71.3	62.5	61.5	66.4	73.6	86.6	94.1	99.2	(21)	
		MCWB	66.9	66.1	64.4	59.7	57.3	54.2	53.3	54.5	56.5	59.3	62.4	64.0	(22)	
	5%	DB	96.1	93.2	88.8	78.6	67.4	60.4	59.1	62.1	68.5	80.2	88.5	92.6	(23)	
		MCWB	65.2	64.9	62.9	58.8	56.0	53.6	52.5	53.1	55.1	58.1	61.7	62.4	(24)	
	10%	DB	89.5	87.6	82.8	73.8	64.2	58.3	57.2	58.9	64.4	74.4	82.4	86.6	(25)	
		MCWB	63.4	63.4	61.6	57.6	55.4	52.5	51.7	52.1	54.1	56.7	60.1	61.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.1	72.3	69.0	65.4	61.6	58.8	56.7	58.4	61.2	64.0	69.2	70.8	(27)	
		MCDB	85.7	94.6	81.4	76.9	68.0	61.2	60.9	66.7	72.9	78.3	82.7	86.5	(28)	
	2%	WB	69.8	69.4	66.7	63.2	59.5	56.0	55.2	56.0	58.5	61.6	66.1	67.2	(29)	
		MCDB	88.4	85.5	83.0	74.5	66.3	60.4	59.2	63.2	68.2	77.7	85.1	84.0	(30)	
	5%	WB	67.5	67.2	65.2	61.2	58.0	54.6	53.7	54.3	56.7	59.7	63.8	65.0	(31)	
		MCDB	89.4	84.8	82.6	73.6	64.2	59.0	57.8	60.3	65.8	75.4	81.8	85.3	(32)	
	10%	WB	65.6	65.4	63.5	59.3	56.5	53.2	52.3	52.8	54.9	57.9	61.4	63.1	(33)	
		MCDB	86.7	82.6	80.2	70.1	62.5	57.5	56.2	58.1	62.7	71.4	78.0	82.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	31.0	30.0	27.9	25.0	18.8	16.9	15.5	18.4	22.2	28.5	30.5	30.7	(35)	
		MCDBR	41.0	37.2	34.7	31.7	22.5	18.5	16.8	22.6	29.8	37.9	39.3	41.0	(36)	
		MCWBR	13.2	12.5	12.5	13.7	11.7	11.7	10.9	13.6	15.6	15.2	14.0	13.6	(37)	
	5% WB	MCDBR	30.6	30.1	28.3	25.2	18.1	15.9	14.5	19.7	24.8	31.2	33.4	30.6	(38)	
		MCWBR	11.9	11.3	11.8	12.7	10.8	11.1	10.4	13.0	14.7	14.3	13.4	11.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.346	0.336	0.320	0.312	0.303	0.298	0.298	0.310	0.335	0.348	0.341	0.335	(40)	
	taud		2.468	2.515	2.552	2.561	2.575	2.576	2.570	2.518	2.437	2.404	2.460	2.496	(41)	
	Ebn at Noon		314	310	303	287	271	263	270	283	292	302	313	318	(42)	
	Edh at Noon		37	34	30	26	23	21	23	27	33	38	37	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2450	2161	1725	1229	830	684	735	1018	1396	1841	2180	2402	(44)	
	RadStd		124	110	73	71	42	47	31	55	112	127	118	108	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (25 neighbors)		+0.79	N/A	N/A	+1.66	N/A	N/A	-29	-265	+302	+85	

Nomenclature: See separate page